

Project Overview

Project Details



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Internship Program: Cyberster Blue Team Internship

Organization: Cyberster

Week 6(Days 35–42)

Date: July 29, 2026

Status: Completed

Executive Overview

During this project, a simulated insider threat attack was performed in a controlled lab environment to demonstrate how sensitive data can be staged, encoded, exfiltrated, and investigated using Security Operations Center (SOC) tools.

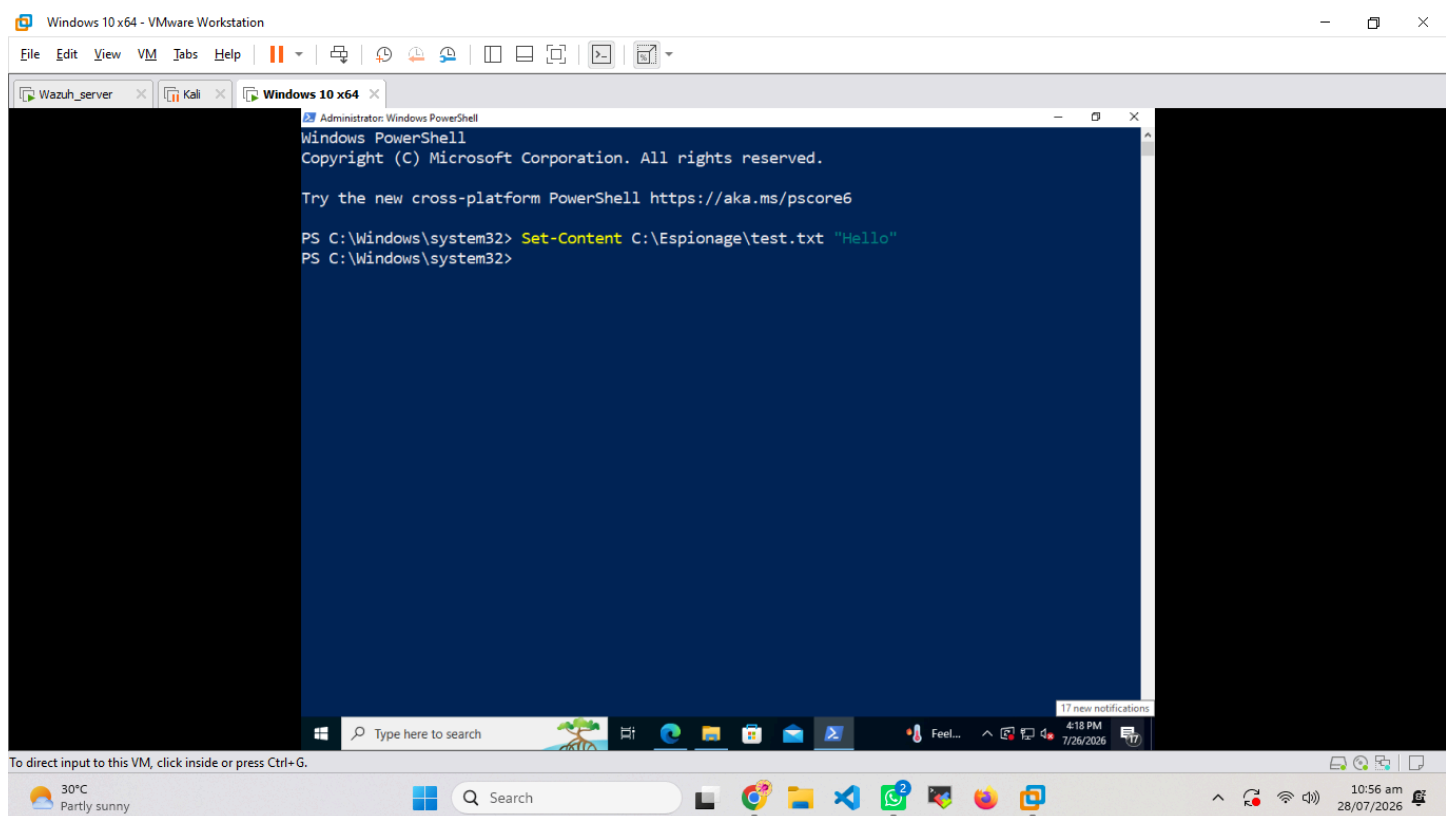
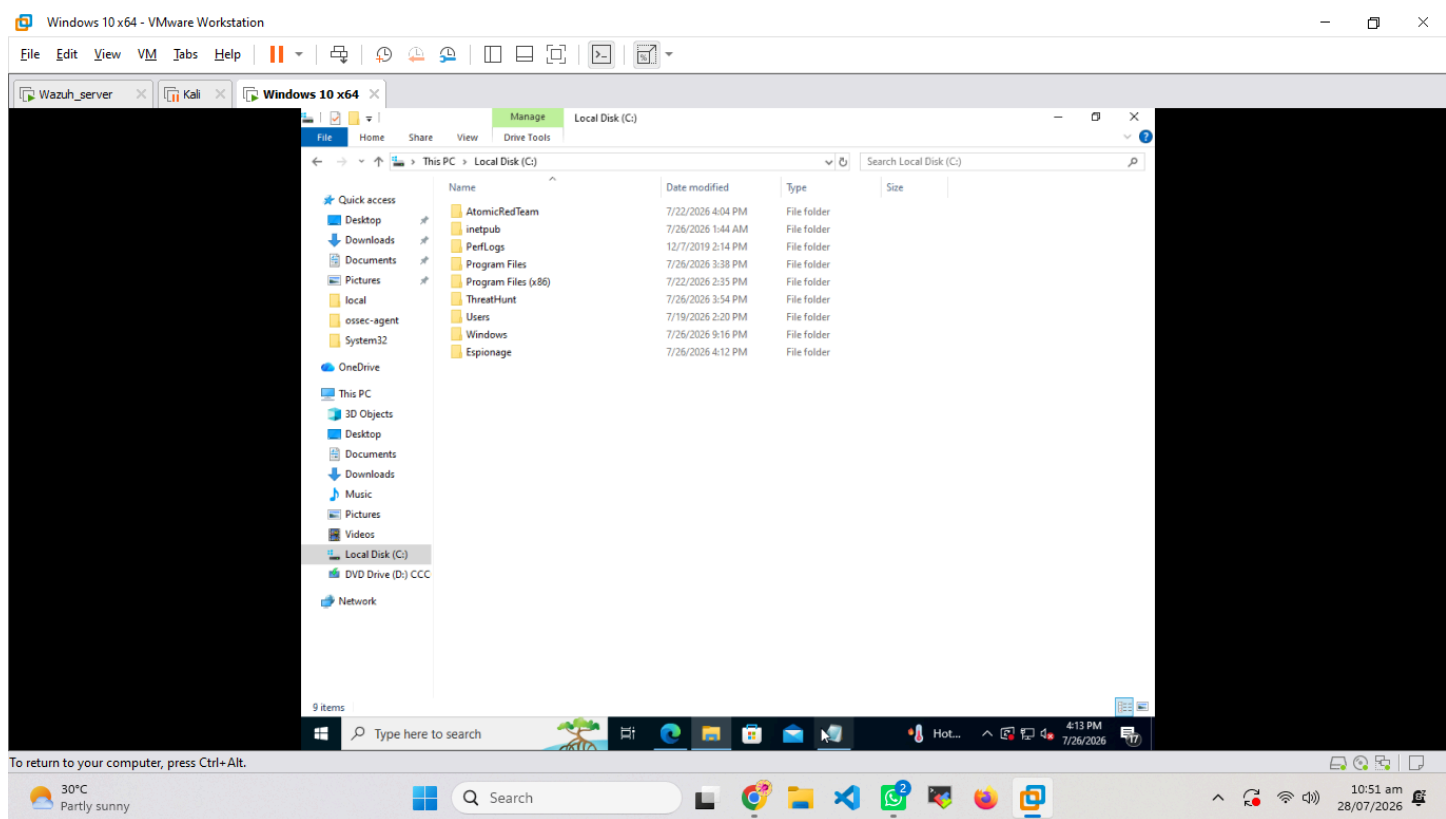
The attack was executed on a Windows 10 endpoint monitored by the Wazuh Agent. A confidential file was created inside the C:\Espionage directory, encoded into Base64 using the Windows built-in certutil.exe utility, and transmitted to a Kali Linux machine using an HTTP POST request. After the exfiltration, the attacker deleted the files in an attempt to remove evidence and avoid detection.

Despite the cleanup attempt, Wazuh File Integrity Monitoring (FIM) successfully recorded the file creation and deletion events. These alerts enabled the complete attack timeline to be reconstructed during the investigation. The captured Base64 payload was also decoded using CyberChef, confirming the contents of the exfiltrated file.

As part of the Detection Engineering phase, a custom Wazuh rule was developed to detect .b64 files created inside the monitored directory. The rule was tested successfully and improved the monitoring capability for similar insider threat activity.

This project demonstrates the complete incident response lifecycle, including attack simulation, log analysis, forensic investigation, detection engineering, and incident documentation.

Detection and Analysis



Windows 10 x64 - VMware Workstation

File Edit View VM Tabs Help

Wazuh_server Kali Windows 10 x64

Manage Local Disk (C:)

File Home Share View Drive Tools

This PC > Local Disk (C:)

Search Local Disk (C:)

Quick access Desktop Downloads Documents Pictures local ossec-agent System32 OneDrive This PC 3D Objects Desktop Downloads Documents Music Pictures Videos Local Disk (C:) DVD Drive (D:) CCC Network

Notepad

File Edit Format View Help

```
<!-- File integrity monitoring -->
<syscheck>

<disabled>no</disabled>

<!-- Frequency that syscheck is executed default every 12 hours -->
<frequency>600</frequency>

<!-- Default files to be monitored. -->
<directories check_all="yes" realtime="yes">C:\Espionage</directories>
<directories recursion_level="0" restrict="regedit.exe|system.ini|win.ini">%WINDIR%</directories>
<directories realtime="yes" report_changes="yes">C:\Windows\System32</directories>
<directories recursion_level="0" restrict="at.exe|attrib.exe|cacs.exe|cmd.exe|even</directories>
<directories recursion_level="0">%WINDIR%\SysNative\drivers\etc</directories>
<directories recursion_level="0" restrict="WMIC.exe">%WINDIR%\SysNative\wbem</director
<directories recursion_level="0" restrict="powershell.exe">%WINDIR%\SysNative\WindowsP
<directories recursion_level="0" restrict="winrm.vbs">%WINDIR%\SysNative</directories>

<!-- 32-bit programs. -->
<directories recursion_level="0" restrict="at.exe|attrib.exe|cacs.exe|cmd.exe|even</directories>
<directories recursion_level="0">%WINDIR%\System32\drivers\etc</directories>
<directories recursion_level="0" restrict="WMIC.exe">%WINDIR%\System32\wbem</director
<directories recursion_level="0" restrict="powershell.exe">%WINDIR%\System32\WindowsP
```

Ln 83, Col 5 100% Windows (CRLF) UTF-8

Sunday, July 26, 2026

To direct input to this VM, click inside or press Ctrl+G.

30°C Partly sunny Search 10:52 am 28/07/2026

Windows 10 x64 - VMware Workstation

File Edit View VM Tabs Help

Wazuh_server Kali Windows 10 x64

Manage Local Disk (C:)

File Home Share View Drive Tools

This PC > Local Disk (C:)

Search Local Disk (C:)

Quick access Name Date modified Type Size File folder AtomicRedTeam 7/22/2026 4:04 PM

Administrator: Command Prompt

```
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>net stop wazuh && net start wazuh

The Wazuh service was stopped successfully.

The Wazuh service was started successfully.

C:\Windows\system32>
```

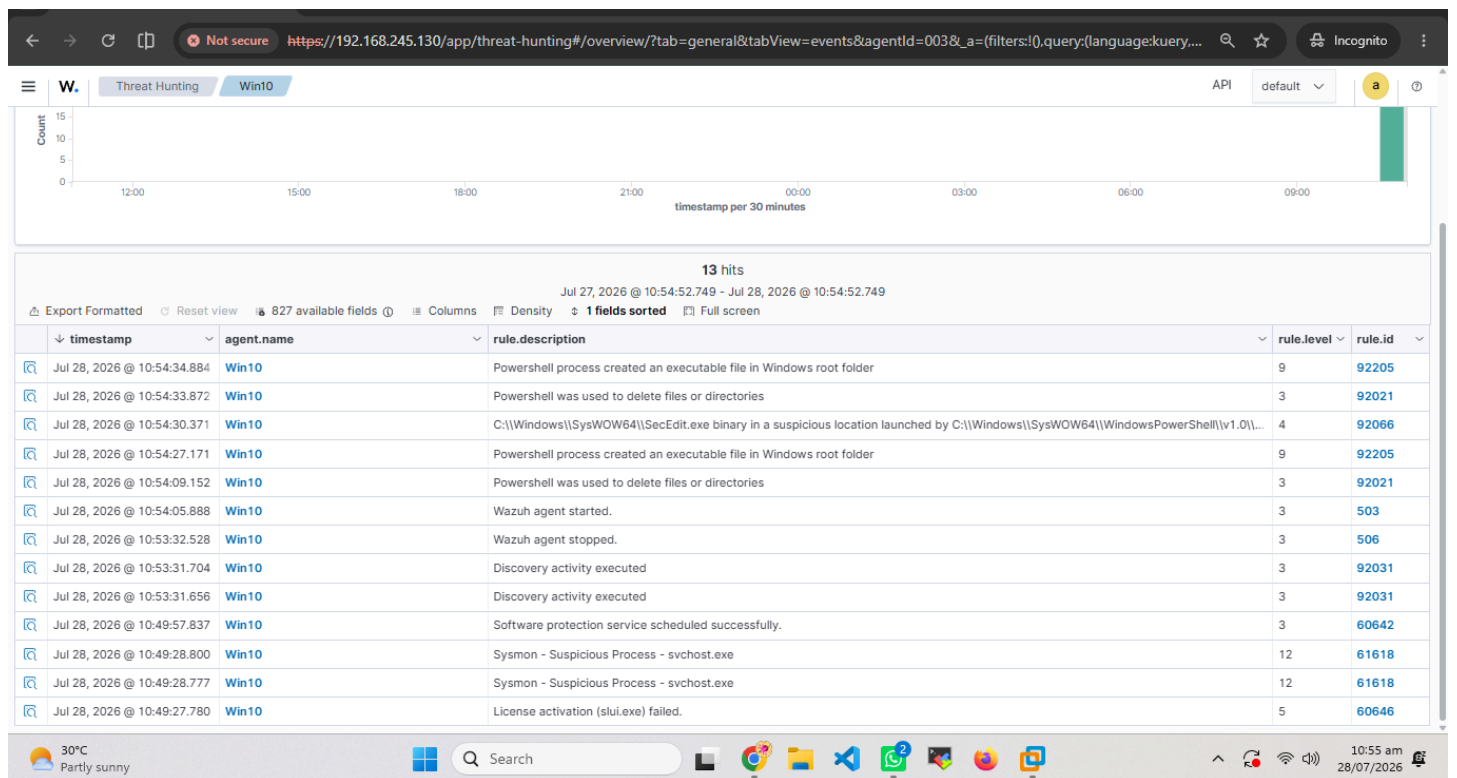
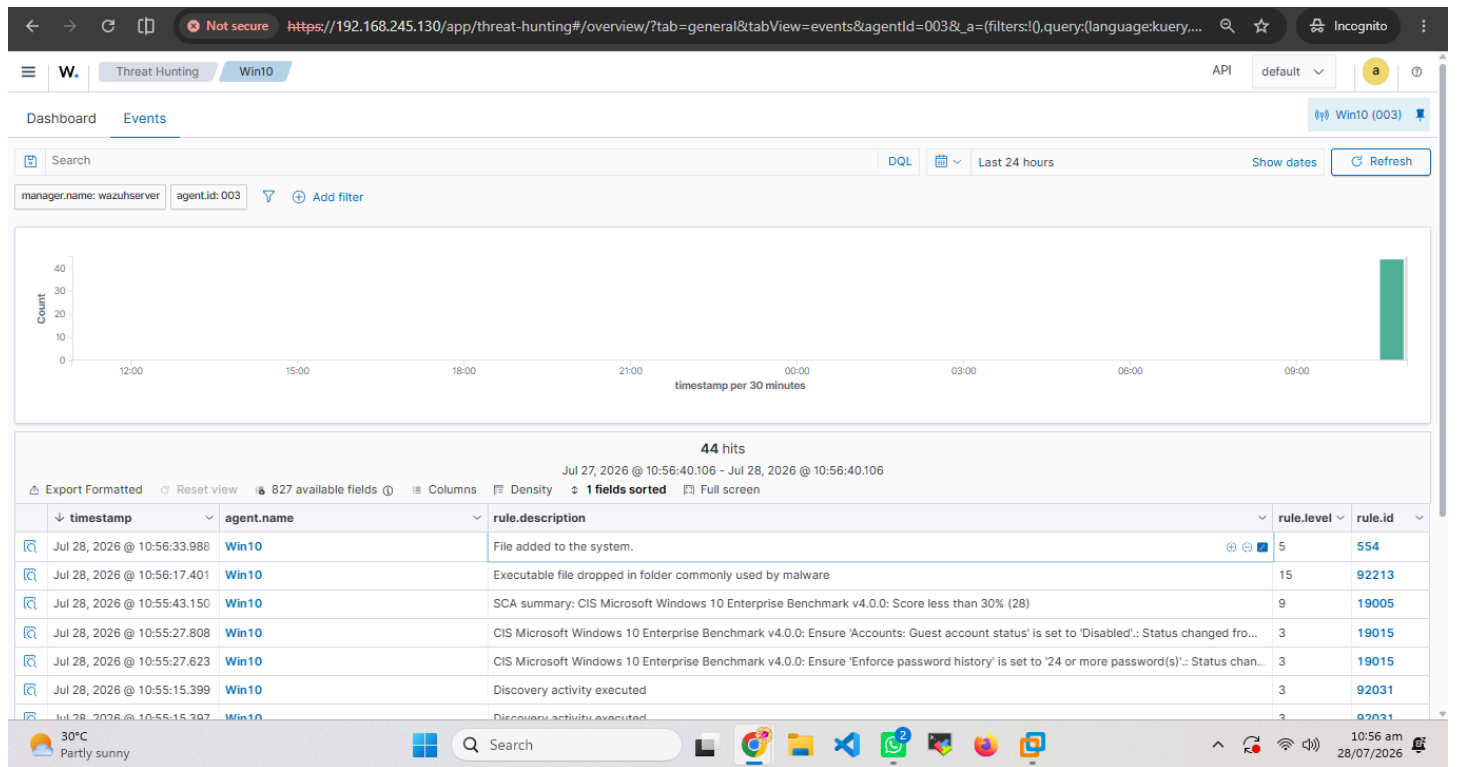
9 items

Type here to search

Sunday, July 26, 2026

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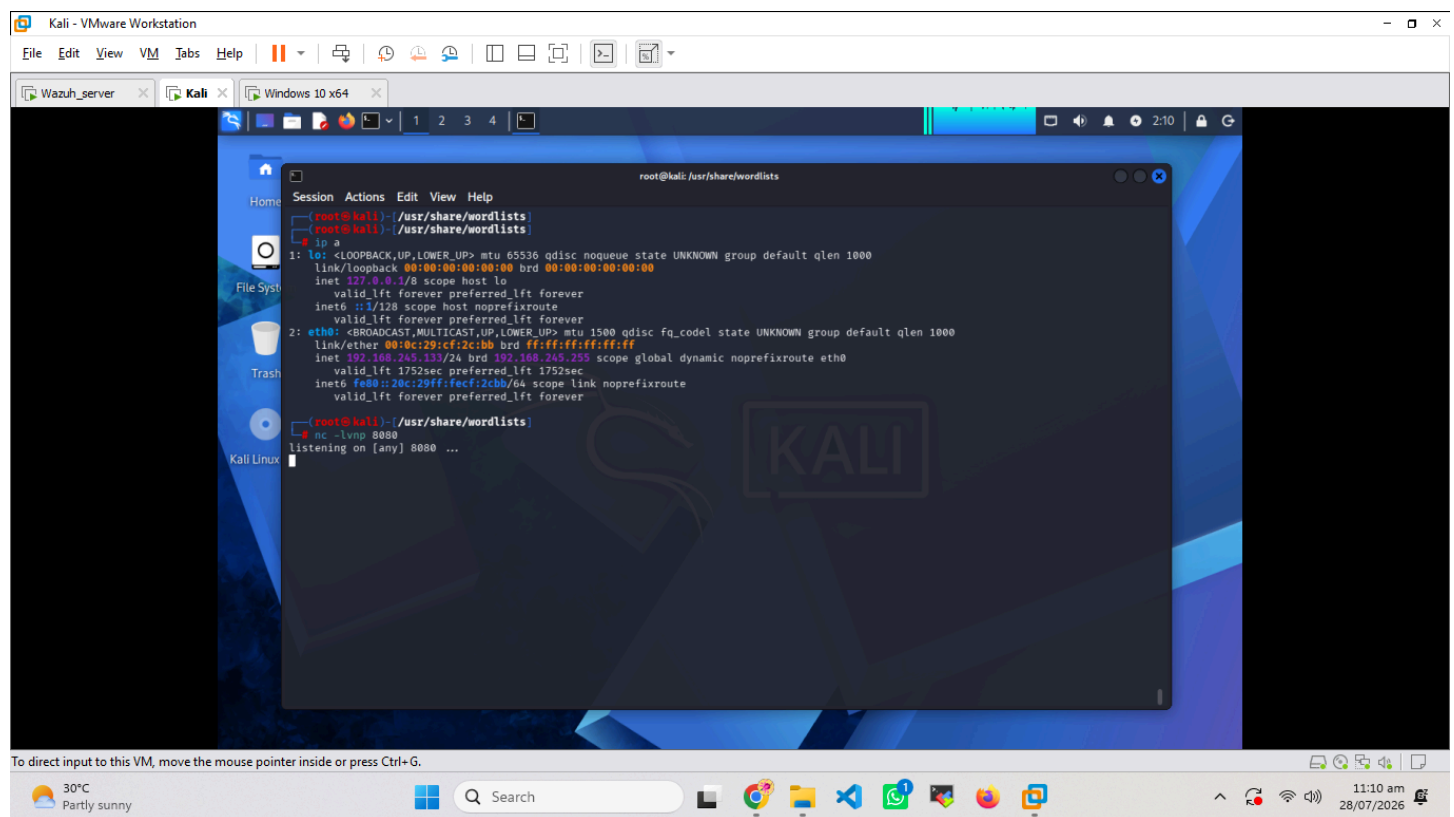
Attack Overview

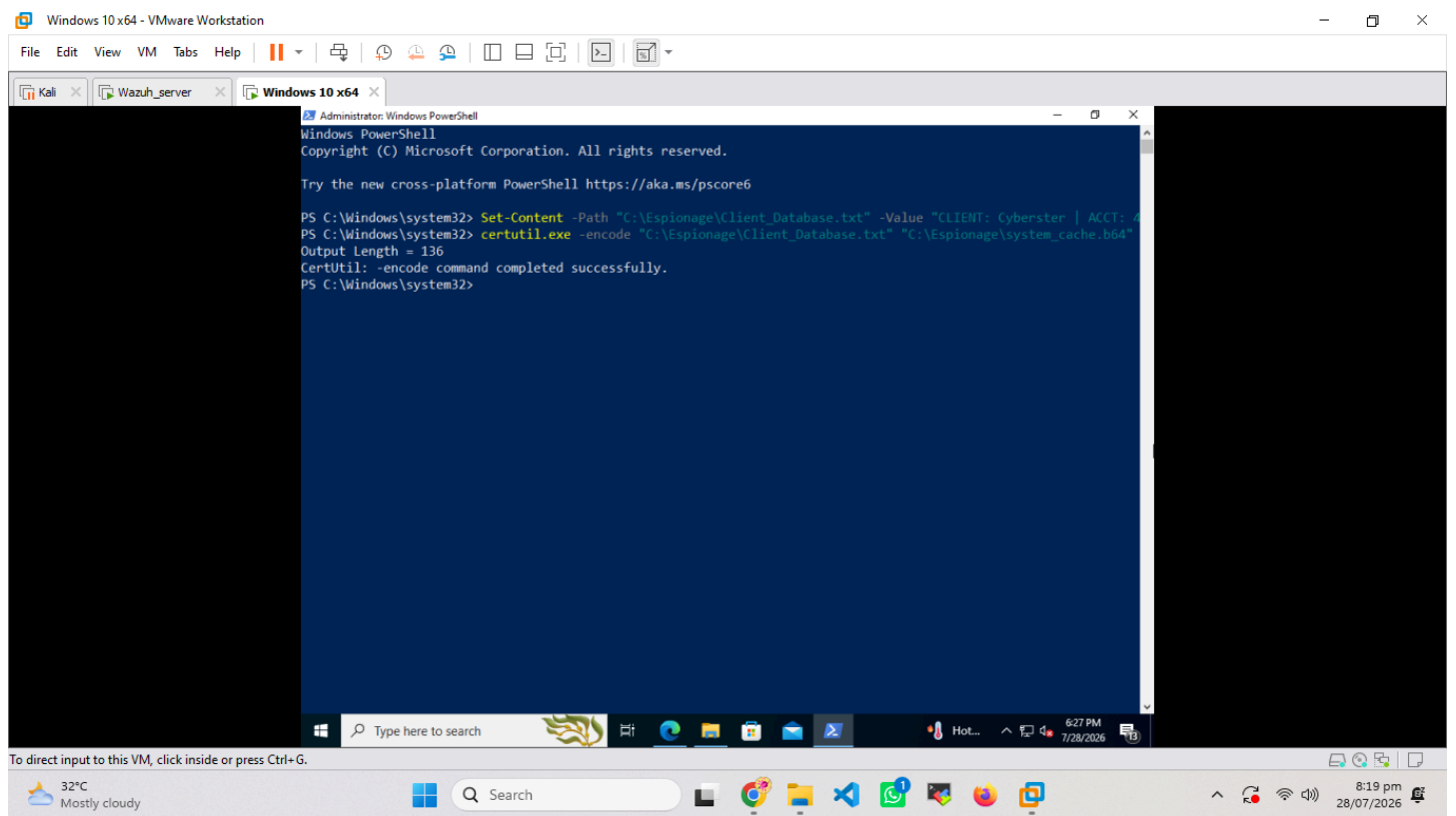
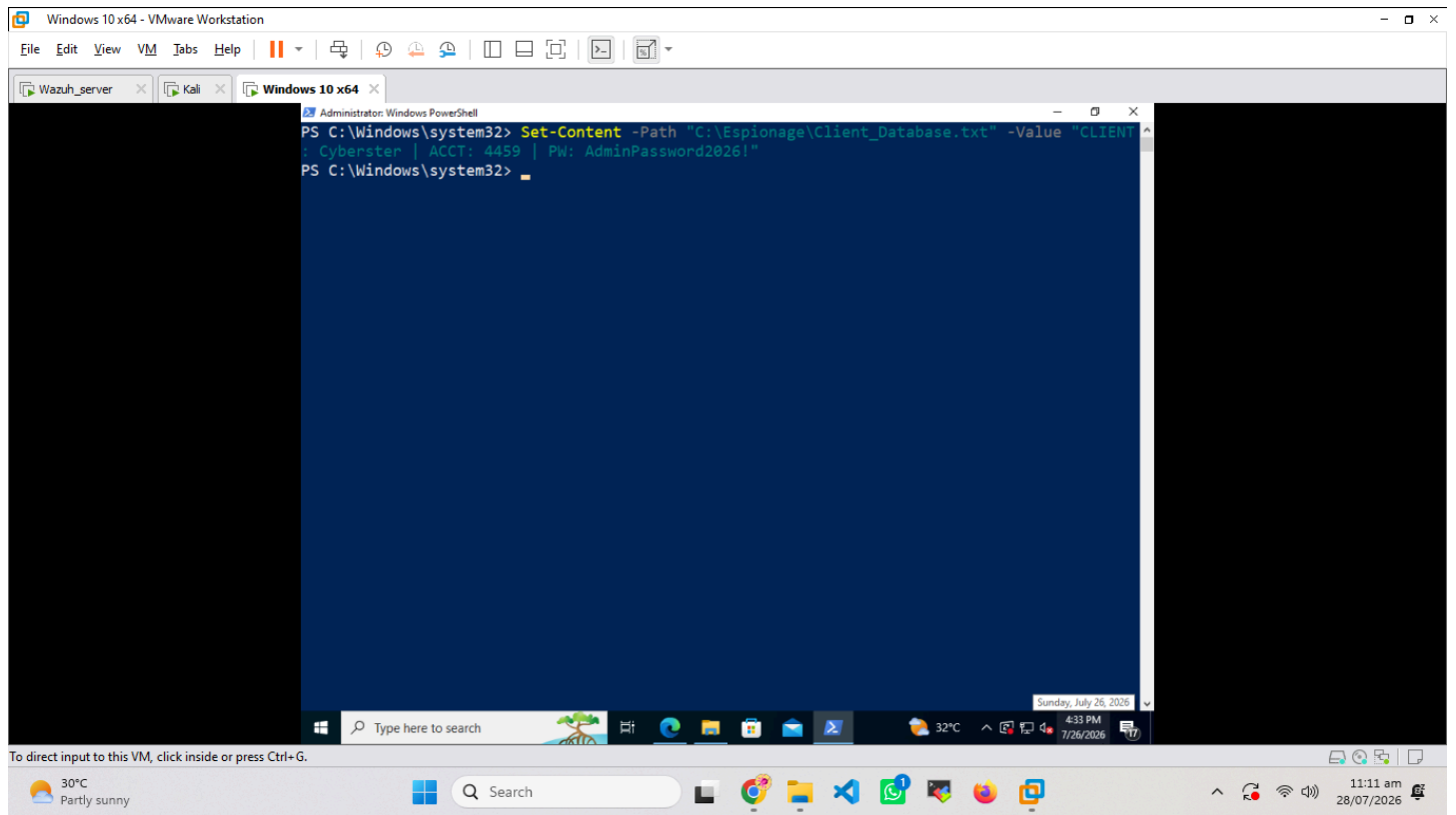
The incident began when an insider with legitimate access created a confidential file on the Windows endpoint. Instead of sending the file directly, the attacker used the Windows `certutil.exe` utility to encode the file into Base64 format. The encoded file was then transmitted to a Kali Linux system using an HTTP POST request. After completing the transfer, both the original and encoded files were deleted in an attempt to hide the attack.

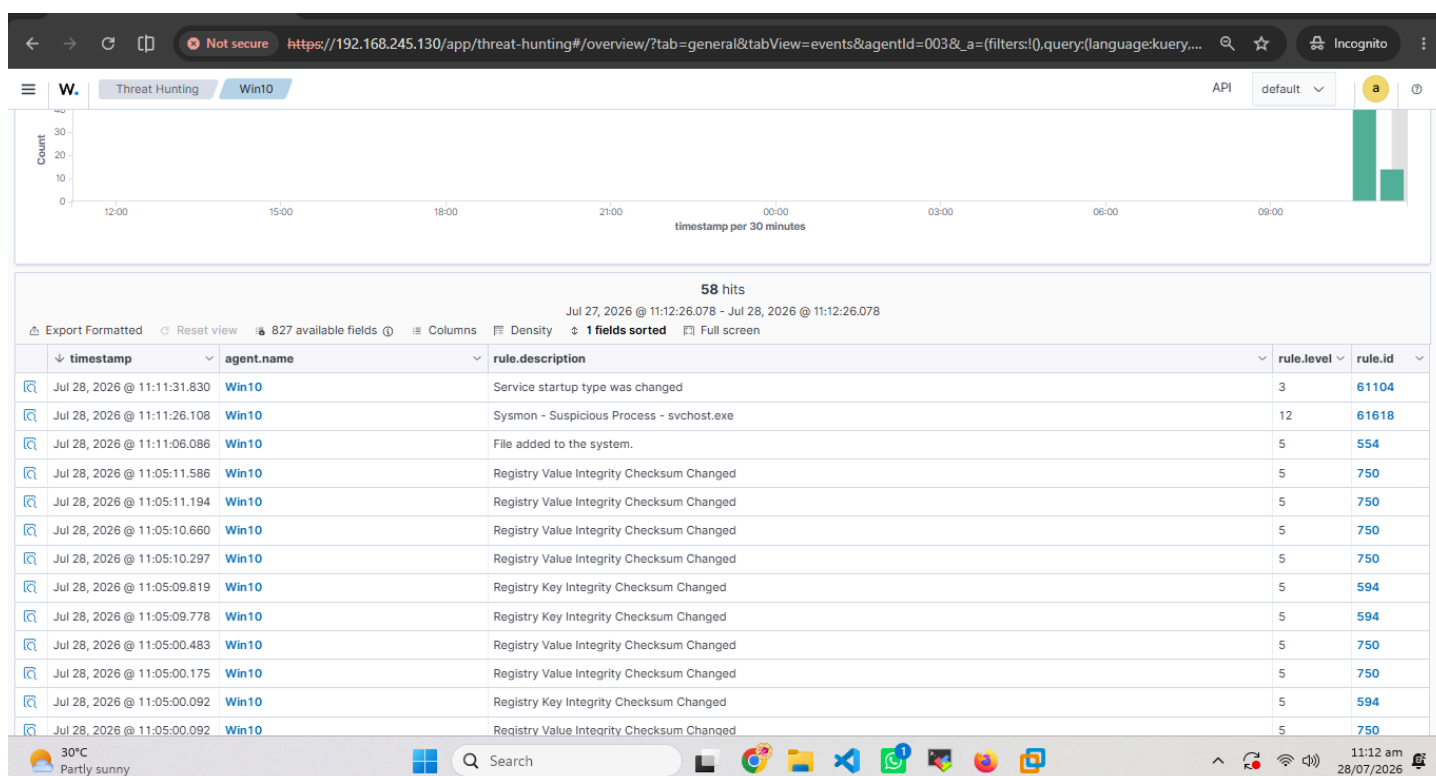
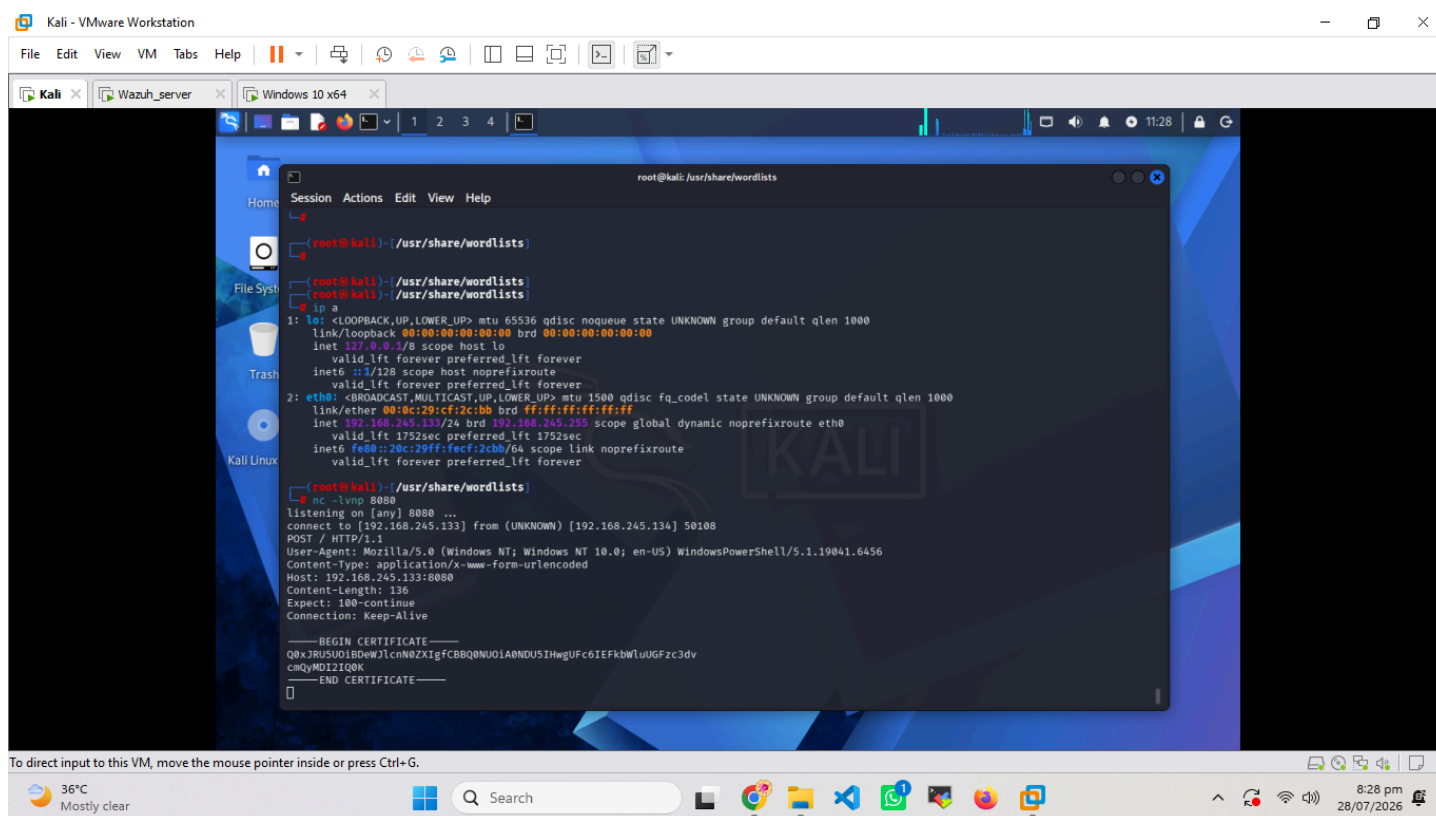
Although the attacker removed the files from the system, the activity had already been recorded by Wazuh File Integrity Monitoring, allowing the attack to be investigated successfully.

4.2 Attack Timeline

Time	Activity
06:00	Created Client_Database.txt
06:05	Created system_cache.b64 using CertUtil
06:10	Exfiltrated Base64 data to Kali Linux
06:15	Deleted both files from the monitored directory





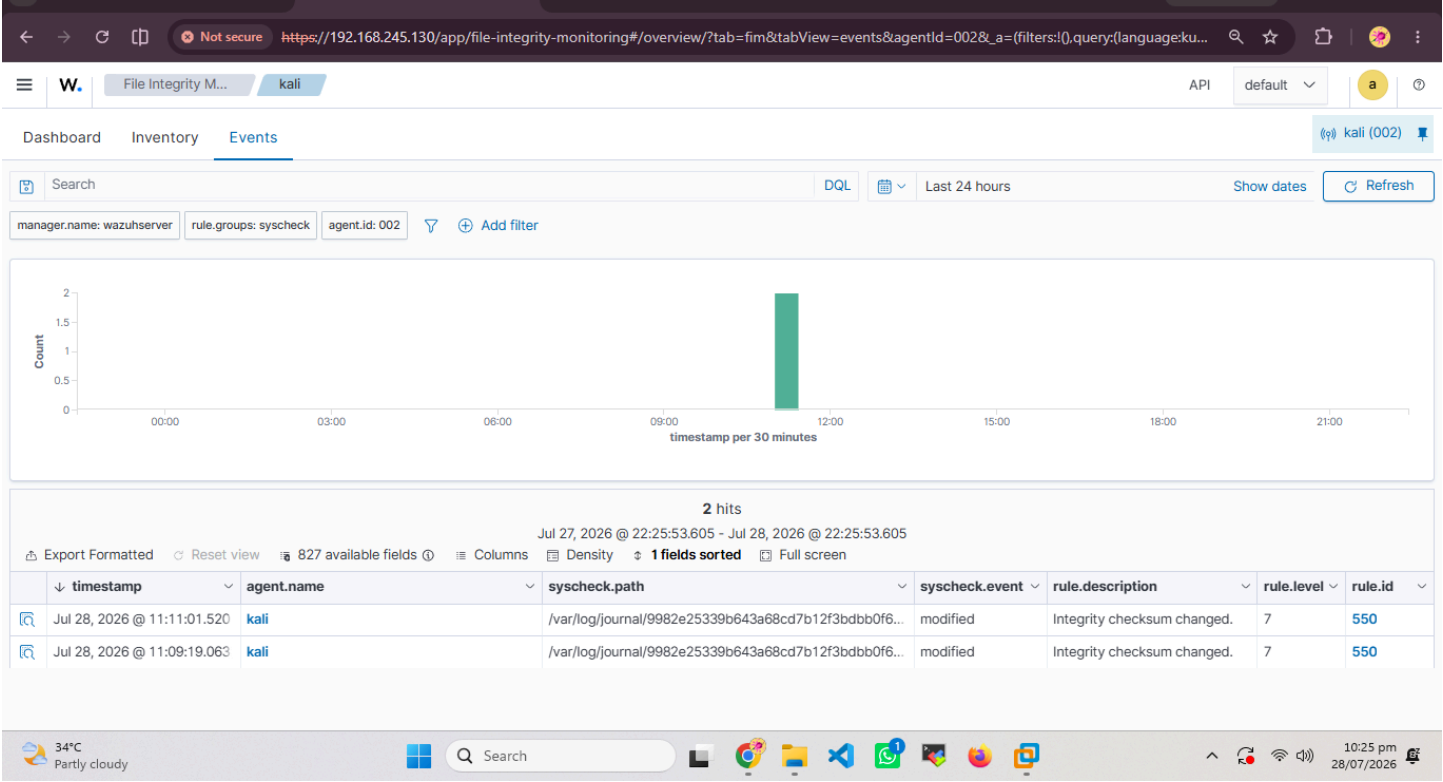


4.3 Wazuh Investigation

The investigation started by reviewing File Integrity Monitoring alerts in the Wazuh Dashboard. The alerts confirmed every stage of the attack and helped reconstruct the complete sequence of events.

The first alerts showed that a confidential file and its Base64-encoded version were created inside the monitored directory. Later alerts confirmed that both files were deleted shortly after the data had been transferred.

These logs provided reliable forensic evidence, even though the attacker attempted to remove all traces of the activity.

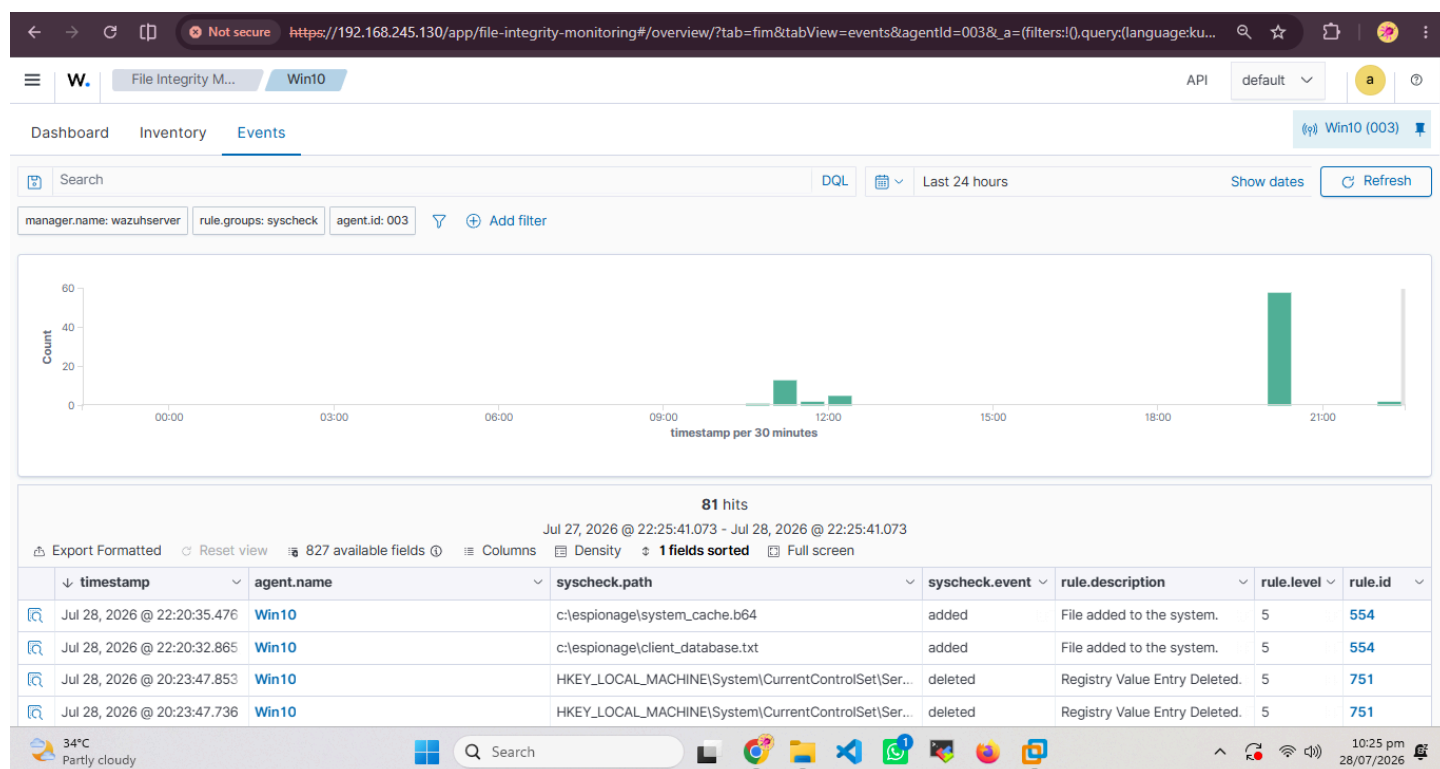


Rule550_Create.png

Figure 1: Wazuh File Integrity Monitoring detected the creation of Client_Database.txt and system_cache.b64 inside the monitored directory.

The investigation continued by reviewing file deletion events.

Wazuh generated additional alerts when both files were removed from the system. This confirmed that the attacker attempted to erase evidence after successfully completing the data exfiltration.



Rule553_Delete.png

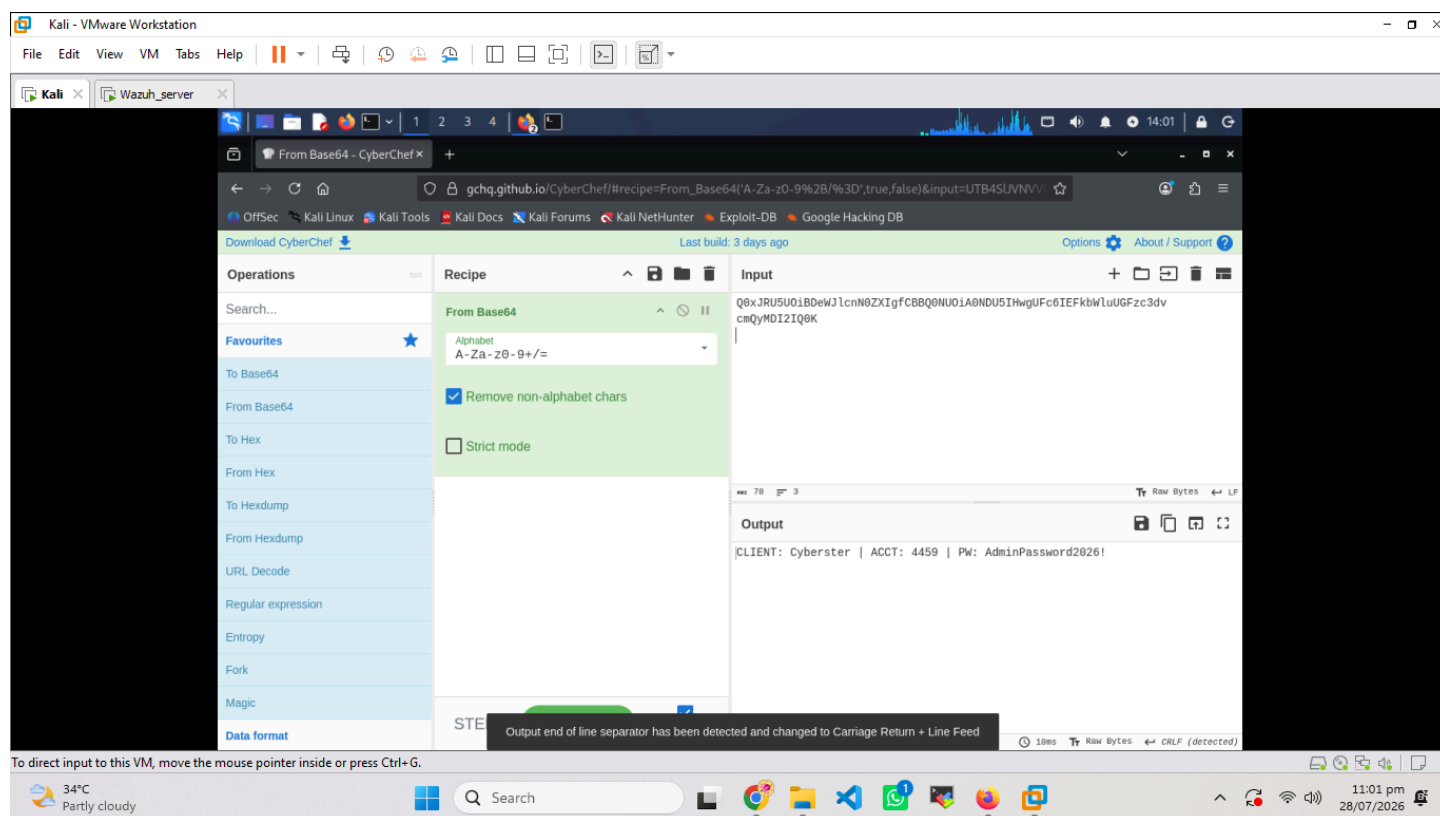
Figure 2: Wazuh detected the deletion of the original and encoded files after the exfiltration phase.

4.4 Evidence Analysis

To verify the contents of the stolen data, the Base64 payload captured by the Kali Linux listener was analyzed using CyberChef.

The payload was decoded successfully, revealing the original confidential information. This confirmed that the attacker had successfully encoded and transferred the intended data before deleting the local files.

The decoded output matched the contents of the original file created during the attack simulation.



CyberChef.png

Figure 3: CyberChef successfully decoded the captured Base64 payload, revealing the original confidential information.

4.5 Indicators of Compromise (IOCs)

The following indicators were identified during the investigation.

IOC Type	Value
Monitored Directory	C:\Espionage
Original File	Client_Database.txt
Encoded File	system_cache.b64
Utility Used	certutil.exe
Exfiltration Method	HTTP POST
Destination	Kali Linux Netcat Listener
Monitoring Solution	Wazuh File Integrity Monitoring

4.6 MITRE ATT&CK Mapping

Technique ID	Technique
T1074	Data Staged
T1027	Obfuscated Files or Information
T1048	Exfiltration Over Alternative Protocol
T1070	Indicator Removal on Host

4.7 Investigation Summary

The investigation confirmed that the insider successfully staged sensitive information, encoded it using Base64, transmitted it to an external system, and attempted to remove the evidence. Wazuh File Integrity Monitoring preserved the necessary forensic artifacts, allowing the complete attack to be reconstructed.

The collected evidence clearly demonstrated the value of continuous file monitoring and centralized logging for detecting insider threats and supporting incident response activities.

Static Malware Analysis

Containment, Eradication and Recovery

5.1 Containment

After confirming the insider threat activity, the first priority was to contain the incident and prevent any further data loss. Although this project was performed in a lab environment, the following actions

represent the appropriate response in a real organization.

The affected Windows endpoint should be isolated from the network to stop any additional communication with external systems. The user account involved in the incident should be temporarily disabled while the investigation is in progress. All security logs and forensic evidence should be preserved to support further analysis and ensure that no important information is lost.

These actions help prevent the attacker from continuing malicious activity while allowing investigators to examine the incident safely.

5.2 Eradication

After containment, the next step is to remove any malicious artifacts and verify that the endpoint is clean.

The suspicious files created during the attack should be removed from the system. Security analysts should review the endpoint for any additional unauthorized files, scheduled tasks, startup entries, or persistence mechanisms that may have been created by the attacker.

The system should also be scanned for signs of compromise, and all unnecessary administrative privileges should be reviewed. Any weaknesses identified during the investigation should be addressed before returning the system to normal operation.

5.3 Recovery

Once the system has been verified as clean, normal operations can be restored.

User access should be re-enabled only after confirming that the endpoint is secure. Passwords associated with the affected account should be reset, and the system should continue to be monitored for any unusual activity.

Additional monitoring through Wazuh should remain active to detect similar behavior in the future. The successful implementation of the custom detection rule also improves the organization's ability to identify suspicious Base64-encoded files at an early stage.

Post-Incident Activity

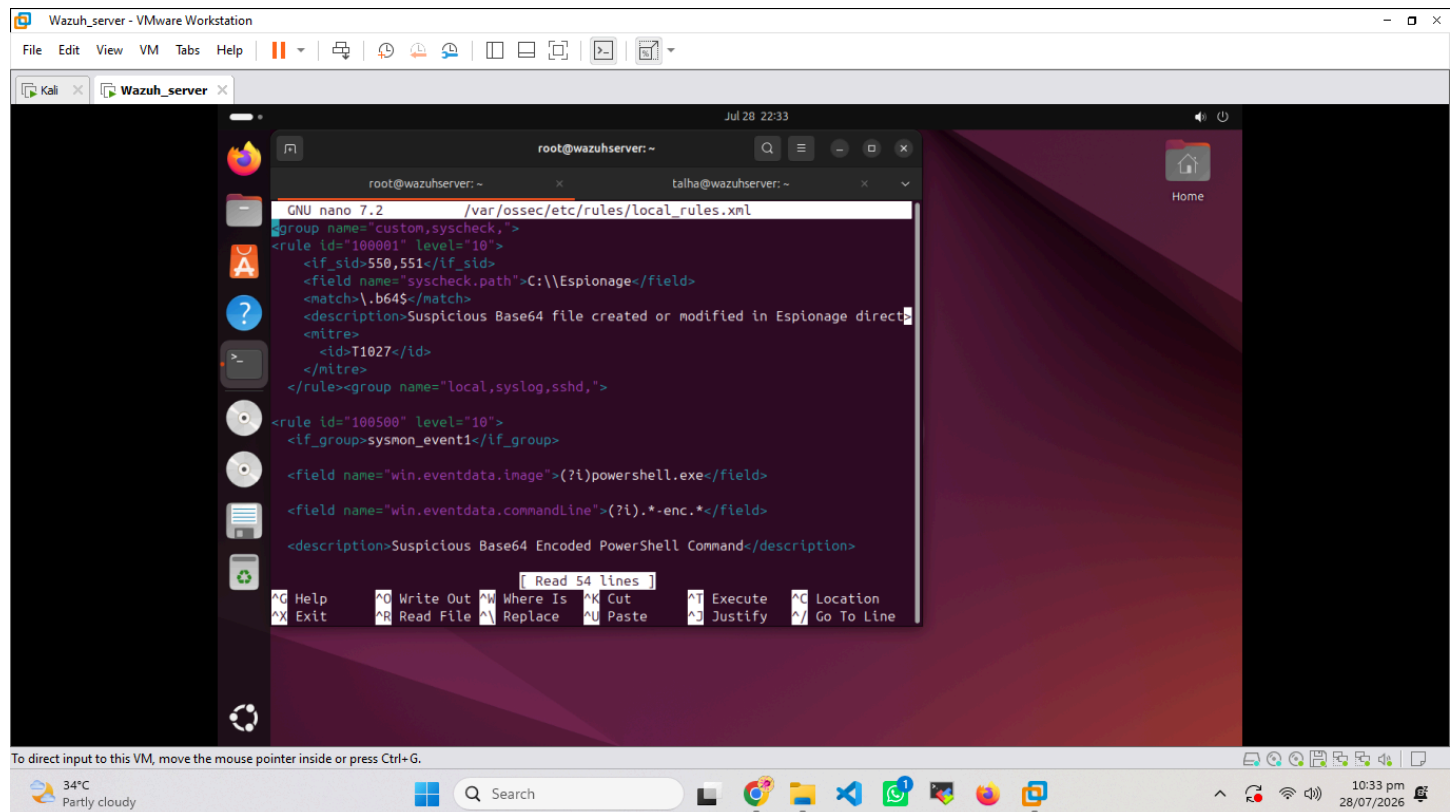
After completing the investigation, additional security improvements were implemented to strengthen the monitoring environment and improve future detection capabilities.

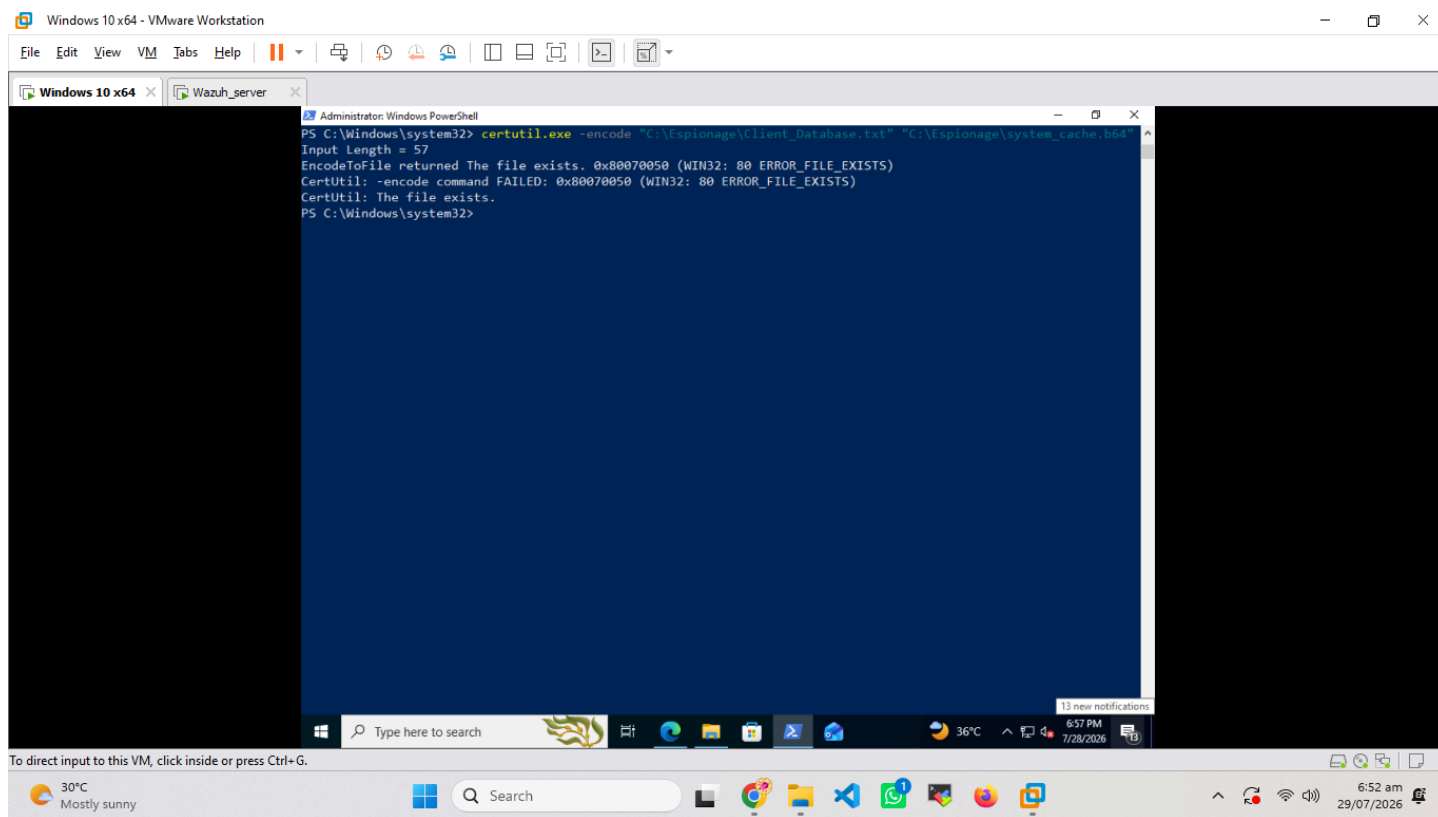
6.1 Detection Engineering

During the investigation, it was observed that the default Wazuh File Integrity Monitoring rules detected general file activity but did not specifically identify Base64-encoded files.

To improve visibility, a custom Wazuh detection rule was created. The rule monitors the C:\Espionage directory and generates a **Level 10** alert whenever a file with the .b64 extension is created or modified.

This enhancement allows security analysts to quickly identify suspicious encoding activity that may indicate data staging or preparation for exfiltration.





Custom_Rule.png

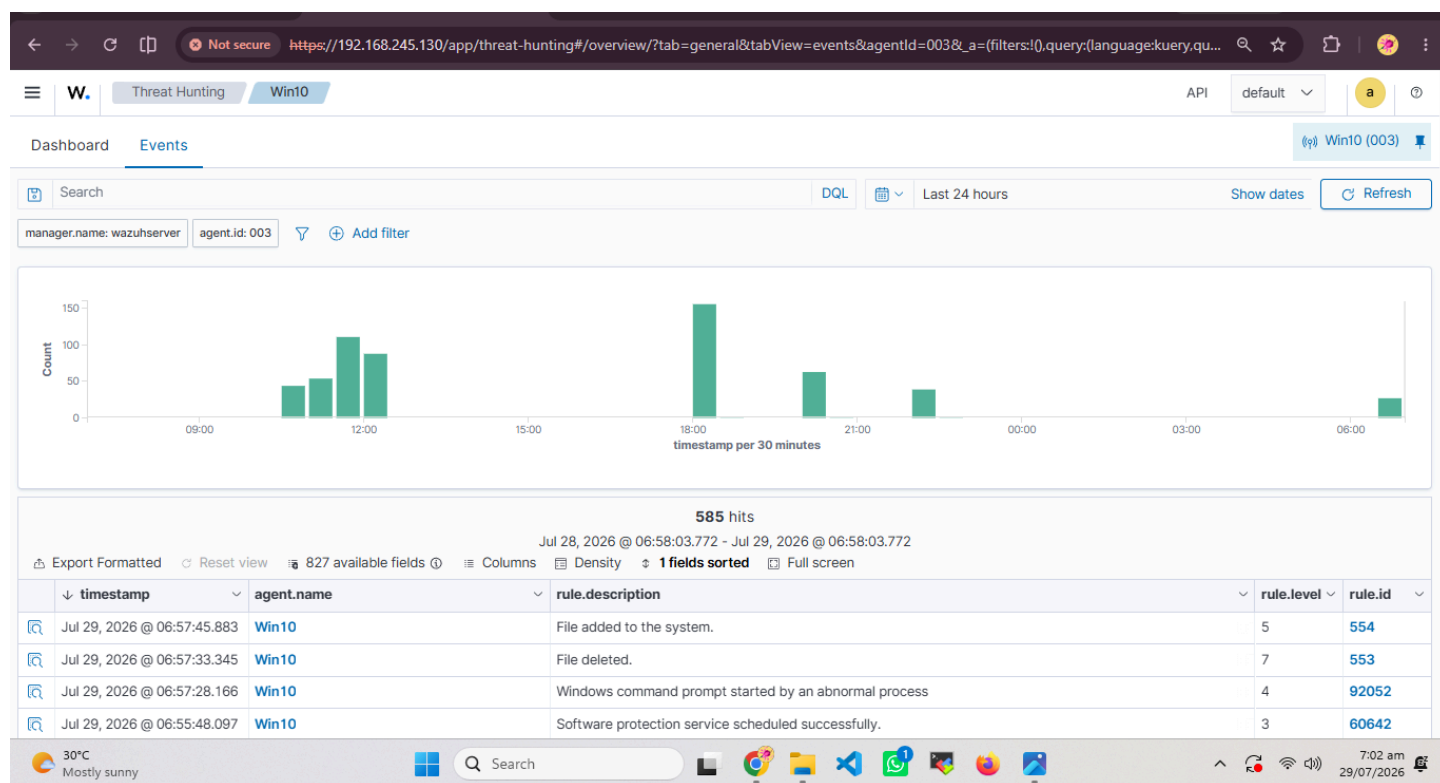
Figure 4: Custom Wazuh detection rule created to detect .b64 files inside the monitored directory.

6.2 Rule Validation

After creating the custom rule, the Wazuh Manager service was restarted successfully. The attack was then partially repeated by generating the Base64 file again.

The custom rule triggered the expected alert in the Wazuh Dashboard, confirming that the detection logic was working correctly.

This validation demonstrated that the monitoring environment had been improved and could detect similar activity more effectively in future incidents.



Custom_Alert.png

Figure 5: Successful validation of the custom Wazuh detection rule after recreating the Base64-encoded file.

6.3 Lessons Learned

This project provided valuable practical experience in both attack simulation and defensive security operations.

Some key lessons learned include:

- File Integrity Monitoring is effective for tracking file activity, even if files are later deleted.
- Centralized logging provides valuable forensic evidence during incident investigations.
- Base64 encoding can be detected by creating custom detection rules.
- Detection Engineering helps improve security monitoring after an incident.
- MITRE ATT&CK mapping provides a structured way to understand attacker behavior.
- Proper incident documentation is essential for effective communication and future improvements.

Conclusion

This project demonstrated the complete lifecycle of an insider threat incident, from attack execution to investigation, detection engineering, and incident response.

Using Wazuh File Integrity Monitoring, the attack was successfully detected and reconstructed even after the attacker attempted to delete the evidence. The investigation confirmed the sequence of events, identified the indicators of compromise, and verified the exfiltrated data through CyberChef analysis.

To strengthen future detection capabilities, a custom Wazuh rule was developed and validated successfully. This improvement increases visibility into suspicious file activity and supports faster incident detection.

Overall, this project provided practical experience in SOC operations, threat hunting, digital forensics, detection engineering, and incident response while demonstrating the importance of continuous monitoring and evidence-based investigations in defending against insider threats.